Tame and Affine

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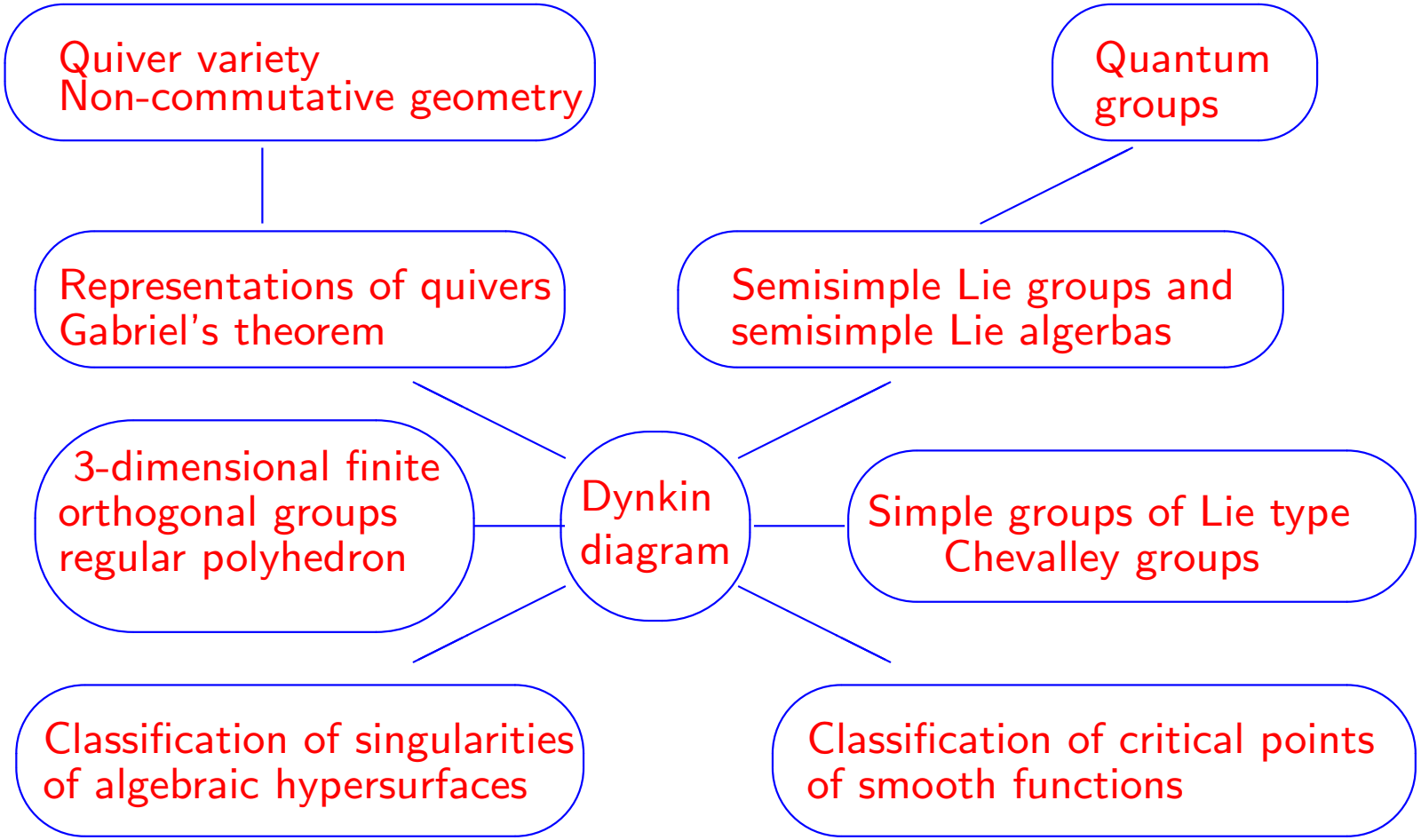
1.Introduction

Consider the following problems



Arnold's problem

寻求所有分类定理的共同根源以及对分类问题的相似性提出先验的证明来代替事后的验证。

There are three well-known researches.

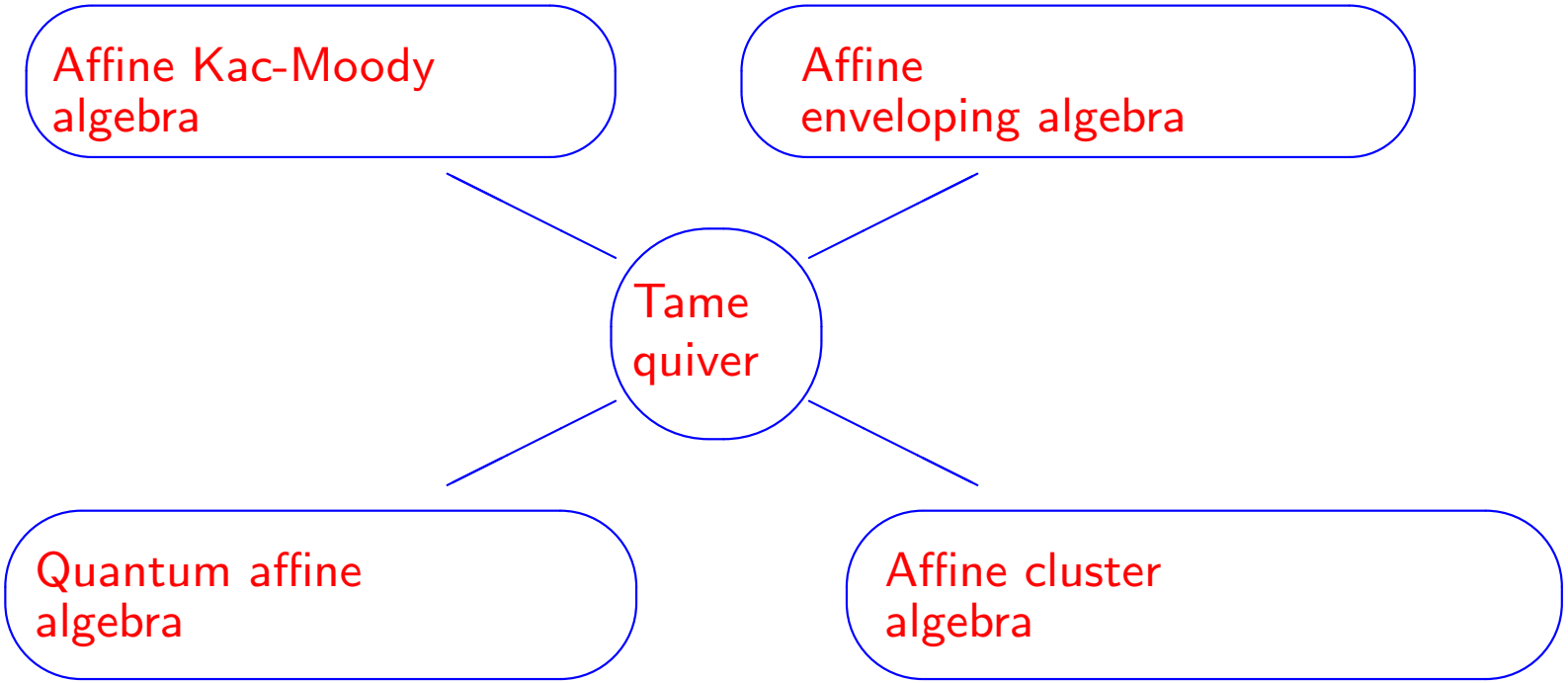
(a) Chevalley

$\longrightarrow$ Loop groups $\longrightarrow$ Kac-Moody groups

(b) Mckay Correspondence

(c) Lie theory from singularity theory: E. Brieskorn and K.Saito

Of course, not only Dynkin diagrams of finite type.
Also, diagrams of affine type and indefinite type.


Affine Kac-Moody algebra
Affine enveloping algebra
Tame quiver
Quantum affine algebra
Affine cluster algebra

Representations of tame quivers

Let Q be a tame quiver without oriented cycles (tame quiver=quiver whose underline graph is extended Dynkin).

- ⋆ Classification of indecomposable modules: preprojective, preinjective and regular modules.
- ⋆ Let $\mathrm{Prep}(Q)$, $\mathrm{Prei}(Q)$ and $\mathrm{Reg}(Q)$ be the full subcategory consisting of all preprojective, preinjective and regular modules.
- ⋆ regular part: tubes indexed by $\mathbb{P}^1(\mathbb{C})$, where there are $s(\leq 3)$ non-homogeneous tubes, others are homogeneous tubes.
- ⋆ non-homogeneous tube $\longleftrightarrow$ nilpotent representations of cyclic quiver; homogeneous tube $\longleftrightarrow$ nilpotent representations of Jordan quiver.

Ringel-Hall algebras

$k = \mathbb{F}_q$, $v = \sqrt{q}$.
A: finite dimensional k-algebra. $\mathrm{Ext}^2(-,-) = 0$
$A = kQ$: the path algebra of quiver Q
$\mathcal{P}$ ={iso-classes of fin-dim A-module}$\supset \mathcal{I}$ ={iso-classes of simple A-module}$= Q_0$

Grothendieck group: $G(A) = \mathbb{Z}\mathcal{I}$

Euler form

$< -, - >: \mathbb{Z}\mathcal{I} \times \mathbb{Z}\mathcal{I} \longrightarrow \mathbb{Z}$

$$< i, j >= \begin{cases} 1 & i = j \\ -\#\{\text{the path from } i \text{ to } j\} & i \neq j \end{cases}$$

Symmetric Euler form: $(i, j) =< i, j > + < j, i >$

For any $\alpha \in \mathcal{P}$, choose $V_\alpha \in \alpha$, let $a_\alpha = \#\mathrm{Aut}_A(V_\alpha)$.
If $\alpha, \beta, \gamma \in \mathcal{P}$, let

$$g^\lambda_{\alpha\beta} = \#\{\text{the submodule } B \text{ of } V_\lambda | B \simeq V_\beta, V_\lambda / B \simeq V_\alpha\}$$

The Ringel-Hall algebra

$$\mathfrak{h}(A) = \bigoplus_{\alpha \in \mathcal{P}} \mathbb{C} u_\alpha$$

with multiplication:

$$u_\alpha u_\beta = v^{<\alpha,\beta>} \sum_{\lambda \in \mathcal{P}} g^\lambda_{\alpha\beta} u_\lambda$$

Here

$$< \alpha, \beta > = < \underline{\dim} V_\alpha, \underline{\dim} V_\beta >$$

$$= \dim_k \mathrm{Hom}_A(V_\alpha, V_\beta) - \dim_k \mathrm{Ext}^1_A(V_\alpha, V_\beta)$$

This is an associative algebra.

Composition algebra

The subalgebra $\mathcal{C}(A)$ of $\mathfrak{h}(A)$ which is generated by $u_i, i \in \mathcal{I}$ is called composition algebra.

Theorem (Ringel 1990,Green 1995)

$\mathcal{C}(A) \simeq U_v^+(\mathfrak{g})$ (quantized enveloping algebra defined by Drinfeld-Jimbo). $\mathfrak{g}$ Kac-Moody Lie algebra corresponding to the Cartan datum $(-,-) : \mathbb{Z}\mathcal{I} \times \mathbb{Z}\mathcal{I} \to \mathbb{Z}$.

Geometric version of Ringel-Hall algebras

Λ: $\mathbb{C}Q/\langle R\rangle$;
$\mathbb{E}_{\underline{d}}(\Lambda)$: the variety of Λ-modules of dimension vector $\underline{d}$;
$G_{\underline{d}} \curvearrowright \mathbb{E}_{\underline{d}}(\Lambda)$.
$\mathcal{O}_i$: $G_{\underline{d}_i}$-invariant constructible subset of $\mathbb{E}_{\underline{d}_i}$ for i=1,2. For $M \in \mathrm{mod}\Lambda$, Put

$$V(\mathcal{O}_1, \mathcal{O}_2; M) = \{0 = M_0 \subseteq M_1 \subseteq M_2 = M \mid M_i/M_{i-1} \in \mathcal{O}_i, i = 1, 2\}$$

$M_{G_{\underline{d}}}(\Lambda)$: the $\mathbb{C}$-space of $G_{\underline{d}}$-invariant constructible function on $\mathbb{E}_{\underline{d}}(\Lambda)$. Define

$$M_G(\Lambda) = \bigoplus_{\underline{d}\in\mathbb{N}^n} M_{G_{\underline{d}}}(\Lambda)$$

with the convolution $1_{\mathcal{O}_1} \bullet 1_{\mathcal{O}_2}(L) = \chi(V(\mathcal{O}_1, \mathcal{O}_2; L))$

2. Tame quivers and affine Lie algebras

Q: tame quiver
X: algebraic variety over $\mathbb{C}$, $F(X)$: the set of all constructible functions, $F_G(X)$: the subspace of $F(X)$ consisting of all G-invariant functions.

$$\mathbf{E}_\alpha = \prod_{h \in \Omega} \mathrm{Hom}_{\mathbb{C}}(\mathbb{C}^{\alpha_{s(h)}}, \mathbb{C}^{\alpha_{t(h)}})$$

$L_\alpha(Q) = F_{G_\alpha}(\mathbf{E}_\alpha)$ for $\alpha \in \mathbb{N}[I]$

$$\mathbf{E}_\alpha \times \mathbf{E}_\beta \xleftarrow{p_1} \mathbf{E}' \xrightarrow{p_2} \mathbf{E}'' \xrightarrow{p_3} \mathbf{E}_{\alpha+\beta}$$

$\mathbf{E}'' = \{(x, W) | x \in \mathbf{E}_{\alpha+\beta}, W$ is $I -$
graded subspace of $\mathbb{C}^{\alpha+\beta}$ such that $x(W) \subset W, \underline{\dim} W = \beta\}$
$\mathbf{E}' = \{(x, W, R^\alpha, R^\beta) | (x, W) \in \mathbf{E}', R^\beta : \mathbb{C}^\beta \simeq W, R^\alpha : \mathbb{C}^\alpha \simeq \mathbb{C}^{\alpha+\beta}/W\}$,
$p_1(x, W, R^\alpha, R^\beta) = (x^\alpha, x^\beta), \quad x_h R^\alpha_{s(h)} = R^\alpha_{t(h)} x^\alpha_h, \;\; x_h R^\beta_{s(h)} = R^\beta_{t(h)} x^\beta_h$.
$p_2(x, W, R^\alpha, R^\beta) = (x, W); \;\; p_3(x, W) = x$.

$$f_\beta * f_\alpha = (p_3)_*(p_2)_b(p_1)^*(g)$$

where $g(x_1, x_2) = f_\alpha(x_1) f_\beta(x_2)$

Denote by $L^{ind}(Q)$ the subspace of $L(Q)$ consisting of all $f \in L(Q)$ such that $f(M) = 0$ if $M \in \operatorname{rep} Q$ is decomposable.
We define $n^*(Q)$ to be the Lie subalgebra of $L^{ind}(Q)$ generated by $\{E_i | i \in I\}$.

$$n^\epsilon(Q) = \sum_{\alpha \in \Delta_+} n^\epsilon_\alpha(Q)$$

$n^\epsilon_\alpha(Q) = \mathbb{C}e_\alpha$ is one dimensional if $\alpha \in \Delta^{re}_+$.
$n^\epsilon_{n\delta}(Q) = \mathbb{C}I/\mathbb{C}\delta$.
The Lie algebra structure of $n^\epsilon(Q)$ is defined using cocycles.

Theorem (Frenkel-Malkin-Vybornov)

- Let Q be a quiver of tame type, then the map $\varphi : n^{\epsilon}(Q) \to n^{*}(Q)$ given by $\varphi(e_i) = E_i$ for $i \in I$ is an isomorphism of Lie algebras.
- The map φ is completely described by the following rule:

$$\varphi(e_\alpha) = \widetilde{E}_\alpha$$

$$\varphi(\alpha_{i,z}(n)) = \widetilde{E}_{i,nN_z,z} - \widetilde{E}_{i+1,nN_z,z}$$

$$\varphi(\alpha_0(n)) = -\widetilde{E}_0(n)$$

for any $\alpha \in \Delta_+^{re}$, $z \in L$ and $i \in \mathbb{Z}/N_z\mathbb{Z}$, $n > 0$.

3.Tame quivers and affine (quantized) enveloping algebras

3.1 An algebraic construction of affine canonical bases

- ⋆ step1: construct a PBW-basis arising from the preprojective, regular and preinjective component respectively
- ⋆ step2: define an (geometric) order on the set of basis elements
- ⋆ step3: there exists a monomial basis whose transition matrix between the PBW-basis is upper-triangular with 1's in the diagonal
- ⋆ step4: slightly modify the PBW-basis (in the regular part) using Schur functions
- ⋆ step5: following Lusztig's method (as in the case of finite type), we obtain the canonical basis

Some notations

Let $\mathcal{M}$ be the set of quadruples $\mathbf{c} = (a_{\mathbf{c}}, b_{\mathbf{c}}, \pi_{\mathbf{c}}, \omega_{\mathbf{c}})$ such that $a_{\mathbf{c}} \in \mathbb{N}_f^{Prep}$ (it corresponds to a preprojective module), $b_{\mathbf{c}} \in \mathbb{N}_f^{Prei}$ (corresponds to a preinjective module),
$\pi_{\mathbf{c}} = (\pi_{1\mathbf{c}}, \pi_{2\mathbf{c}}, \cdots, \pi_{s\mathbf{c}}) \in \Pi_1^a \times \Pi_2^a \times \cdots \Pi_s^a$ is an n-tuple of partitions, and $\omega_{\mathbf{c}} = (\omega_1 \geq \omega_2 \geq \cdots \geq \omega_t)$ is a partition.
Then for each $\mathbf{c} \in \mathcal{M}$ we define

$$E^{\mathbf{c}} = \langle M(a_{\mathbf{c}}) \rangle * E_{\pi_{1\mathbf{c}}} * E_{\pi_{2\mathbf{c}}} * \cdots * E_{\pi_{s\mathbf{c}}} * E_{\omega_{\mathbf{c}}\delta} * \langle M(b_{\mathbf{c}}) \rangle,$$

PBW-basis

Let $\mathcal{A} = \mathbb{Q}[v, v^{-1}]$ and $\mathcal{C}^*(\Lambda)_{\mathcal{A}}$ be the $\mathcal{A}$-subalgebra of the generic composition algebra $\mathcal{C}^*(\Lambda)$ generated by $u_i^{*m} = \frac{u_i^{*m}}{[m]!}$, $i \in I$.

Proposition (Lin-Xiao-Zhang)

The set $\{E^{\mathbf{c}} | \mathbf{c} \in \mathcal{M}\}$ is an $\mathcal{A}$-basis of $U_{\mathbb{Z}}^{+}$.

A geometric order

- For $A \subset \mathbf{E}_\alpha$, $B \subset \mathbf{E}_\beta$, define $A \star B = \{z \in \mathbf{E}_{\alpha+\beta} | \exists$ exact sequence $0 \to M(x) \to M(z) \to M(y) \to 0$ with $x \in B, y \in A\}$
- For $\mathbf{c} \in \mathcal{M}$, define the variety

$$\mathcal{O}_{\mathbf{c}} = \mathcal{O}_{M(\mathbf{a_c})} \star \mathcal{O}_{M_{\pi_{1\mathbf{c}}}} \star \mathcal{O}_{M_{\pi_{2\mathbf{c}}}} \star \cdots \star \mathcal{O}_{M_{\pi_{s\mathbf{c}}}} \star \mathcal{N}_{\omega_{\mathbf{c}}\delta} \star \mathcal{O}_{M(\mathbf{b_c})}$$

where $\mathcal{N}_{\omega_{\mathbf{c}}\delta} = \mathcal{N}_{\omega_1\delta} \star \cdots \star \mathcal{N}_{\omega_t\delta}$ if $\omega_{\mathbf{c}} = (\omega_1, \omega_2, \cdots, \omega_t)$ and $\mathcal{N}_{\omega_i\delta}$ is the union of orbits of regular modules of $\mathfrak{C}(P, L)$ (the embedding of representation category of Kronecker quiver) with dimension vector $\omega_i\delta$.

Bar-invariant basis

Proposition

For any $E^{\mathbf{c}}$, $\mathbf{c} \in \mathcal{M}$, there exists a monomial $\mathfrak{m}_{\mathbf{c}}$ of the divided powers of u_{S_i}, $i \in I$, such that

$$\mathfrak{m}_{\mathbf{c}} = E^{\mathbf{c}} + \sum_{\mathbf{c}' \in \mathcal{M}, \dim \mathcal{O}_{\mathbf{c}'} < \dim \mathcal{O}_{\mathbf{c}}} h^{\mathbf{c}}_{\mathbf{c}'} E^{\mathbf{c}'}$$

where $h^{\mathbf{c}}_{\mathbf{c}'} \in \mathbb{Q}[v, v^{-1}]$.

Theorem (Lin-Xiao-Zhang)

By Lusztig's algebraic method we obtain a bar-invariant basis $\{\mathcal{E}^{\mathbf{c}} | \mathbf{c} \in \mathcal{M}\}$ of $\mathcal{C}^*(\Lambda)_{\mathcal{A}}$. And

$$\mathcal{E}^{\mathbf{c}} = E^{\mathbf{c}} + \sum_{\mathbf{c}' \in \mathcal{M} : \dim \mathcal{O}_{\mathbf{c}'} < \dim \mathcal{O}_{\mathbf{c}}} \zeta^{\mathbf{c}}_{\mathbf{c}'} E^{\mathbf{c}'}$$

where $\zeta^{\mathbf{c}}_{\mathbf{c}'} \in v^{-1}\mathbb{Q}[v^{-1}]$.

Modification of the PBW-basis

inner product on $\mathcal{H}^*(\Lambda)$

$$(\langle M\rangle, \langle N\rangle) = \delta_{MN}\frac{v^{2\mathrm{dimEnd}(M)}}{a_M}$$

Let $\mathcal{L} = \{x \in \mathcal{C}^*(\Lambda) | (x,x) \in v^{-1}\mathbb{Q}[[v^{-1}]] \cap \mathbb{Q}(v)\}$.

The key step is to use $E'_{n\delta}$ instead of $E_{n\delta}$ (which is obtained by certain Schimidt orthogonalization).

Let $P_{n\delta} = nE'_{n\delta}$ and $P_{\omega\delta} = P_{1\delta}^{*r_1} * \cdots * P_{t\delta}^{*r_t}$ for a partition $\omega = (1^{r_1}2^{r_2}\cdots t^{r_t})$. Then let $S_{\omega\delta}$ be the Schur functions corresponding to $P_{\omega\delta}$, and for $\mathbf{c} \in \mathcal{M}$, define

$$e^{\mathbf{c}} = \langle M(\mathbf{a_c})\rangle * E_{\pi_{1\mathbf{c}}} * E_{\pi_{2\mathbf{c}}} * E_{\pi_{s\mathbf{c}}} * S_{\omega_{\mathbf{c}}} * \langle M(\mathbf{b_c})\rangle$$

Affine canonical basis

Proposition

$\{e^{\mathbf{c}} | \mathbf{c} \in \mathcal{M}\}$ is an almost othogonal basis of $\mathcal{L}$, i.e. $(e^{\mathbf{c}}, e^{\mathbf{c}'}) \in \delta_{\mathbf{c}\mathbf{c}'} + v^{-1}\mathbb{Q}[[v^{-1}]] \cap \mathbb{Q}(v)$.

Theorem, Lin-Xiao-Zhang

Again by applying Lusztig's algebraic method to $\{e^{\mathbf{c}} | \mathbf{c} \in \mathcal{M}\}$, we obtain an $\mathcal{A}$-basis of $\mathcal{C}^*(\Lambda)_{\mathcal{A}}$ which is characterized by the following properties:

- $\star$ $\bar{\mathcal{E}'^{\mathbf{c}}} = \mathcal{E}'^{\mathbf{c}}$
- $\star$ $\pi(\mathcal{E}'^{\mathbf{c}}) = \pi(\mathcal{E}^{\mathbf{c}})$, where $\pi : \mathcal{L} \to \mathcal{L}/v^{-1}\mathcal{L}$ is the canonical projection.
- $\star$ $(\mathcal{E}'^{\mathbf{c}}, \mathcal{E}'^{\mathbf{c}'}) \equiv \delta_{\mathbf{c}\mathbf{c}'} \pmod{v^{-1}\mathbb{Q}[[v^{-1}]] \cap \mathbb{Q}(v)}$

And finally, the basis $\mathcal{E}'^{\mathbf{c}} | \mathbf{c} \in \mathcal{M}$ is the canonical basis in the sense of Lusztig.

3.2 $\mathbb{Z}$-bases of the affine enveloping algebras

- ⋆ using methods of Lin-Xiao-Zhang, we can obtain a $\mathbb{C}$-basis (PBW-type) of the affine enveloping algebra U^+
- ⋆ but it is difficult to prove that this basis is a $\mathbb{Z}$-basis of the $\mathbb{Z}$-form
- ⋆ we modify the basis elements arising from homogeneous tubes, the idea comes from Macdonald's theory of symmetric functions
- ⋆ then we obtain an integral basis of U^+

Notations

We use the geometric definition of Hall algebras over $\mathbb{C}$.
Let $\mathcal{J}$ be the set of all quadruples $\mathbf{c} = (P_{\mathbf{c}}, I_{\mathbf{c}}, \pi_{\mathbf{c}}, \lambda_{\mathbf{c}})$ where $P_{\mathbf{c}} \in \mathrm{Prep}(Q)$ is a preprojective module, $I_{\mathbf{c}} \in \mathrm{Prei}(Q)$ is a preinjective module, $\pi_{\mathbf{c}} = (\pi_{\mathbf{c}1}, \pi_{\mathbf{c}2}, \cdots, \pi_{\mathbf{c}s})$ is an n-tuple of partitions and $\omega_{\mathbf{c}} \vdash n, n \in \mathbb{N}$.

For any $\mathbf{c} \in \mathcal{J}$, we define

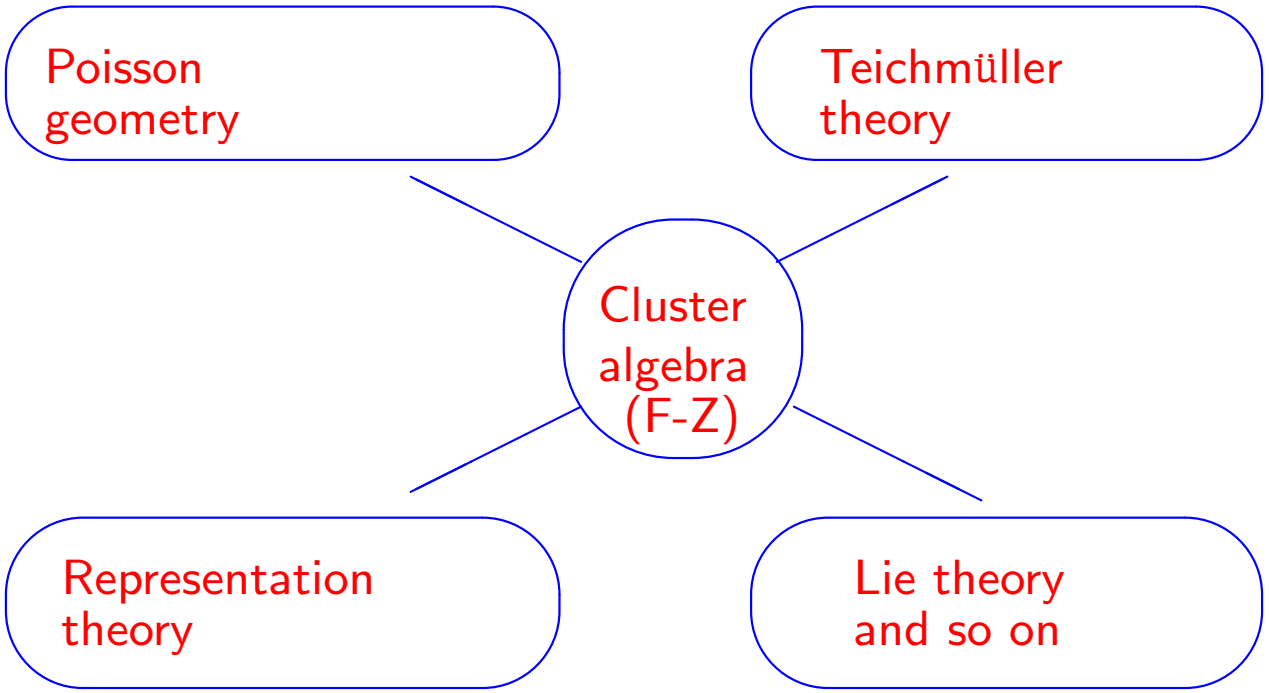
$$B_{\mathbf{c}} = \mathbf{1}_{P_{\mathbf{c}}} E_{\pi_{\mathbf{c}1},1} E_{\pi_{\mathbf{c}2},2} \cdots E_{\pi_{\mathbf{c}s},s} M_{\omega_{\mathbf{c}}\delta} \mathbf{1}_{I_{\mathbf{c}}},$$

Here $\mathbf{1}_{P_{\mathbf{c}}}$ is the characteristic function of the orbit of the module $P_{\mathbf{c}}$ (the same meaning for $\mathbf{1}_{I_{\mathbf{c}}}$), $E_{\pi_{\mathbf{c}i},i}$ is the basis element arising from non-homogeneous tubes (similar to Lin-Xiao-Zhang), $M_{\omega_{\mathbf{c}}\delta}$ is the basis element arising from the embedding of the representation category of Kronecker quiver (and this part is different from the quantum case).

Theorem (Jiang-Sheng)

The set $\{B_{\mathbf{c}} | \mathbf{c} \in \mathcal{J}\}$ is a $\mathbb{Z}$-basis of the integral composition algebra $\mathcal{C}_{\mathbb{Z}}(Q)$ $(= U_{\mathbb{Z}}^{+})$.

4. Tame quivers and affine cluster algebras



- Description: cluster algebra is a commutative algebra (in $\mathbb{Q}(x_1, \cdots, x_n)$) endowed with a distinguished system of generators (cluster variables) grouped into overlapping subsets (clusters) which are constructed recursively via mutation.
- Aim: providing an algebraic framework for the study of total positivity in canonical bases of quantum groups.

An example: A_2

$$
\begin{tikzcd}
 & x_2 \arrow[r, "x_1"] & x_2 \arrow[dr] & & \xrightarrow{x_2} \\
x_1 \arrow[ur] & & & x_1' & \\
 & x_2' \arrow[r, "x_1'"] & x_2' \arrow[dr] & & \xrightarrow{x_2'} \\
x_1' \arrow[ur] & & & x_1'' & \\
 & x_2'' \arrow[r, "x_1''"] & x_2'' \arrow[dr] & & \cdots \\
x_1'' \arrow[ur] & & & x_1''' &
\end{tikzcd}
$$

$$x_1' = \frac{1+x_2}{x_1},\ x_2' = \frac{1+x_1'}{x_2} = \frac{1+x_1+x_2}{x_1x_2},\ x_1'' = \frac{1+x_1}{x_2},\ x_2'' = x_1,\ x_1''' = x_2$$

- clusters: $\{x_1, x_2\},\ \{x_2, x_1'\},\ \{x_1', x_2'\}\ \cdots$
- cluster variables: $\{x_1, x_2, x_1', x_2', x_1''\}$
- cluster algebra: generated by cluster variables

Cluster category (A. Buan, R. Marsh, M. Reineke, I. Reiten and G. Todorov)

The cluster category associated to Q is the orbit category

$$\mathcal{C}(Q) := \mathcal{D}^b(Q)/F$$

where $F = T \circ \tau^{-1}$, the shift functor T and AR-translation τ.

- (B. Keller) $\mathcal{C}(Q)$ is triangulated category.
- Any $M \in \mathrm{obj}\mathcal{C}(Q)$ has the decomposition $M_0 \oplus TP$ (up to isomorphism) where $M_0, P \in \mathrm{mod}\mathbb{C}Q$ and P projective.

The Caldero-Chapoton map

1. if M is an indecomposable $\mathbb{C}Q$-module, then

$$X_M^Q = \sum_{\mathbf{e}} \chi(\mathrm{Gr}_{\underline{e}}(M)) \prod_{i \in Q_0} x_i^{-\langle \underline{e}, s_i \rangle - \langle s_i, \underline{\dim} M - \underline{e} \rangle};$$

2. if $M = TP_i$ is the shift of the projective module associated to $i \in Q_0$, then

$$X_M^Q = x_i;$$

3. for any two objects M, N of $\mathcal{C}_Q$, we have

$$X_{M \oplus N}^Q = X_M^Q X_N^Q.$$

Where $\langle -, - \rangle$ is the Euler form and $Gr_{\underline{e}}(M)$ is the $\underline{e}$-Grassmannian of M, i.e. the variety of submodules of M with dimension vector $\underline{e}$. X_M^Q is called the generalized cluster variable.

- (P. Caldero and B. Keller) If Q is an acyclic quiver, then

$$\{\text{cluster variables}\} \xrightarrow{1-1} \{\text{indec exceptional object in } \mathcal{C}(Q)\}$$

Cluster multiplication formulas

Let Q be an acyclic quiver and $A = \mathbb{C}Q$.

1. The Caldero-Chapton's formula

Let Q be an acyclic quiver and M any indecomposable non-projective A-module, then

$$X_M X_{\tau M} = 1 + X_E$$

where E is the middle term of the Auslander-Reiten sequence ending in M.

2. P. Caldero and B. Keller's formula

Let Q be an acyclic quiver and M,N be any two objects in $\mathcal{C}(Q)$ such that $dim\ Ext^1_{\mathcal{C}(Q)}(M,N) = 1$, then

$$X_M X_N = X_B + X_{B'}$$

where B and B' are the unique objects such that there exists non-split triangles

$$M \longrightarrow B \longrightarrow N \longrightarrow M[1] \ and \ N \longrightarrow B' \longrightarrow M \longrightarrow N[1]$$

3. J. Xiao and F. Xu's formula

Let Q be an acyclic quiver and M,N be any two objects in $\mathcal{C}(Q)$. Assume that $\mathrm{Ext}_{\mathcal{C}(Q)}(M,N) \neq 0$. Then we have

$$\chi(\mathbb{P}\mathrm{Ext}_{\mathcal{C}(Q)}(M,N))X_M X_N$$

$$= \sum_{B \in \mathcal{Y}} [\chi(\mathbb{P}\mathrm{Ext}_{\mathcal{C}(Q)}(M,N)_{\langle B \rangle}) + \chi(\mathbb{P}\mathrm{Ext}_{\mathcal{C}(Q)}(N,M)_{\langle B \rangle})]X_B$$

where,

$\mathcal{Y}$ is a finite set,

$\langle B \rangle = \{B' \in \mathcal{C}(Q) \mid \chi(Gr_{\underline{e}}(B_0)) = \chi(Gr_{\underline{e}}(B'_0))$ for any $\underline{e} \in \mathbb{N}^{Q_0}\}$

and

$\mathrm{Ext}_{\mathcal{C}(Q)}(M,N)_{\langle B \rangle}$ is the set of equivalence classes of extensions with the middle terms in $\langle B \rangle$.

Notations

- Let $\mathcal{AH}(Q)$ be the subalgebra of $\mathbb{Q}(x_1, \cdots, x_n)$ generated by

$$\{X_M, X_{\tau P} \mid M, P \in \mathrm{mod} - \mathbb{C}Q, P \text{ is projective module}\}.$$

- Let $\mathcal{EH}(Q)$ be the subalgebra of $\mathcal{AH}(Q)$ generated by

$$\{X_M \mid M \in \mathrm{ind} - \mathcal{C}(Q), \mathrm{Ext}^1_{\mathcal{C}(Q)}(M, M) = 0\ \}.$$

A partition of $\mathbb{Z}^{Q_0}$

Let Q be a tame quiver. Define the sets

$$\mathbf{D}(Q) = \{\underline{d} \in \mathbb{N}^{Q_0} \mid \exists \text{regular module } T \oplus R, \underline{\dim}(T \oplus R) = \underline{d}, T \text{ indec.},$$

$$\mathrm{Ext}^1_{\mathcal{C}(Q)}(T, T) \neq 0, \mathrm{Ext}^1_{\mathcal{C}(Q)}(T, R) = \mathrm{Ext}^1_{\mathcal{C}(Q)}(R, R) = 0\}.$$

$\mathbf{E}(Q) = \{\underline{d} \in \mathbb{Z}^{Q_0} \mid \exists L \in \mathcal{C}(Q), \underline{\dim} L = \underline{d}$ and $\mathrm{Ext}^1_{\mathcal{C}(Q)}(L, L) = 0\}$.
Here, we set

$$\underline{\dim}\, TP_i = -\underline{\dim} S_i$$

for $i \in Q_0$. Then we have a partition

$$\mathbb{Z}^{Q_0} = \mathbf{D}(Q) \bigsqcup \mathbf{E}(Q).$$

Fix an assignment (induced by the partition)

$$\phi : \mathbb{Z}^{Q_0} \to \mathrm{obj}(\mathcal{C}(Q)).$$

Define $X_{\underline{d}} := X_{\phi(\underline{d})}$.

The assignment is not unique.
For $\underline{d} \in \mathbb{Z}^{Q_0}$, define $\underline{d}^+ = (d_i^+) = (max(d_i, 0))_{i \in Q_0}$ and $\underline{d}^- = (d_i^-) = \underline{d}^+ - \underline{d}$. Set $P_{\underline{d}^-} = \oplus_{i \in Q_0} P_i^{d_i^-}$. For $\underline{d} \in \mathbb{N}^{Q_0}$, $\exists$ a canonical decomposition $\underline{d} = n\delta + \underline{d}_0$ with $\underline{d}_0 < \delta$.
A canonical assignment is defined as follows.
For $\underline{d} \in \mathbf{E}(Q)$, $\exists M \oplus TP_{\underline{d}^-}$ with M rigid and $\underline{\dim} M = \underline{d}^+$, set $\phi(\underline{d}) = M \oplus TP_{\underline{d}^-}$.
For $\underline{d} \in \mathbf{D}(Q)$, $\exists E[n] \oplus R$ with indecomposable regular simple E and regular rigid R such that $\underline{\dim} E[n] \oplus R = \underline{d}$, set $\phi(\underline{d}) = E[n] \oplus R$.

Bases of cluster algebras of affine type

Theorem-I (M.Ding, J.Xiao and F.Xu)

Let Q be a tame quiver. Then the set $\{X_{\underline{d}} \mid \underline{d} \in \mathbb{Z}^{Q_0}\}$ is a $\mathbb{Z}$-basis of $\mathcal{AH}(Q)$.

- Corollary. $\mathcal{EH}(Q) = \mathcal{AH}(Q)$

Theorem-II (M.Ding, J.Xiao and F.Xu; G. Dupont)

Let Q be a tame quiver. For $\underline{d} \in \mathbb{Z}^{Q_0}$, define $X_{\underline{d}^+} = X_M$ for some $M \in \mathrm{mod}\mathbb{C}Q$ with $\langle M \rangle$ dense in $\mathbb{E}_{\underline{d}^+}$, $X_{\underline{d}^-} = x^{\underline{d}^-}$ and $X_{\underline{d}} = X_{\underline{d}^+} X_{\underline{d}^-}$. Then the set $\{X_{\underline{d}} \mid \underline{d} \in \mathbb{Z}^{Q_0}\}$ is a $\mathbb{Z}$-basis of $\mathcal{AH}(Q)$, called the *generic basis* of $\mathcal{EH}(Q)$.

Quantum cluster algebras

K: a field with $|K| = q$,
$\widetilde{Q}$: an acyclic quiver with m vertices,
$Q \subset \widetilde{Q}$: the full subquiver of n vertices.
Define $\widetilde{B} = (b_{ij})_{m\times n}$ and $\widetilde{R} = (r_{ij})_{m\times n}$ by

$$b_{ij} = |\{\text{arrows}\, i \longrightarrow j\}| - |\{\text{arrows}\, j \longrightarrow i\}|$$

and $r_{ij} = |\{\text{arrows}\, j \longrightarrow i\}|$ for $1 \le i \le m$, $1 \le j \le n$.
Assume that $\Lambda = \Lambda_{m\times m}$ satisfies

$$\Lambda(-\widetilde{B}) = \widetilde{I} := \begin{bmatrix} I_n \\ 0 \end{bmatrix}. \tag{1}$$

$\mathcal{A}_{|K|}(Q) = \mathcal{A}_{|K|}(\Lambda, B)$: the specialized quantum cluster algebra of Q attached to (Λ, B).

The multiplication theorem

The quantum Caldero-Chapoton formula

For a $\mathbb{C}Q$-module M and a projective $\mathbb{C}\widetilde{Q}$-module P, define

$$X_{M\oplus TP} = \sum_{\underline{e}} |Gr_{\underline{e}}M| q^{-\frac{1}{2}\langle \underline{e}, \underline{m}-\underline{e}\rangle} X^{\widetilde{B}\underline{e}-(\widetilde{I}-\widetilde{R})\underline{m}+\underline{\dim} top(P)}.$$

Theorem-I (M.Ding and F.Xu)

Let M and N be kQ-modules. Then

$$q^{[M,N]^1} X_N X_M = q^{-\frac{1}{2}\Lambda((\widetilde{I}-\widetilde{R})\underline{m},(\widetilde{I}-\widetilde{R})\underline{n})} \sum_E \varepsilon_{MN}^E X_E.$$

The multiplication theorem-II

Theorem-II (F.Qin; M.Ding and F.Xu)

Assume that $\mathrm{Hom}_{k\widetilde{Q}}(D_0, \tau A_0 \oplus I) = \mathrm{Hom}_{k\widetilde{Q}}(A_0, I) = 0$. Then $X_N X_M$ is equal to

$$q^{\frac{1}{2}\Lambda((\widetilde{I}-\widetilde{R})\underline{n},(\widetilde{I}-\widetilde{R})\underline{m})} X_E + q^{\frac{1}{2}\Lambda((\widetilde{I}-\widetilde{R})\underline{n},(\widetilde{I}-\widetilde{R})\underline{m})+\frac{1}{2}\langle M,N\rangle-\frac{1}{2}\langle A_0,D_0\rangle} X_{D_0\oplus A_0\oplus I[-1]}.$$

By Theorem-II, $\mathcal{A}(Q)$ is generated by

$$\{X_M | M \text{ is indecomposable rigid } kQ\text{-mod}\} \cup$$

$$\{X_{\tau P_i}, 1 \leq i \leq n | P_i \text{ is indecomposable projective } k\widetilde{Q}\text{-mod}\}.$$

The bases of affine quantum cluster algebras

In the following, assume that Q be either finite or tame quiver.

- If $\underline{d} \in \mathbf{D}(Q)$, define $X_{\underline{d}} := (X_E)^n X_R$ for a regular rigid module R and a regular simple module E.
- If $\underline{d} \in \mathbf{E}(Q)$, define $X_{\underline{d}} := X_M$ for some $M = M_0 \oplus P[1]$ with M_0 rigid.

Theorem (The quantum version of generic bases)

The set

$$\mathcal{B}(Q) := \{X_{\underline{d}} | \underline{d} \in \mathbb{Z}^{Q_0}\}$$

is a $\mathbb{QP}$-basis of $\mathcal{A}_{|k|}(\widetilde{Q}) \otimes_{\mathbb{ZP}} \mathbb{QP}$.

The algebraic Mckay correspondence

Let $S = \mathbb{C}[X, Y]$ and $G \subseteq SL_2(\mathbb{C})$ a finite subgroup. Then the invariant subrings $R = S^G$ of S is a hypersurface $\mathbb{C}[x, y, z]/(f)$ as a Kleininan singularity. Note: R is a direct summand of S as an R-modules.
The algebraic Mckay correspondence is

$$\mathrm{MCM}(R) = \mathrm{add}(S) = \mathrm{proj}(End_R(S)) = \mathrm{proj}(S * G) = \mathrm{proj}(\Lambda_{\widetilde{Q}})$$

where $\mathrm{MCM}(R)$ the (Frobenius) category of Maxmimal Cohen-Macaulay modules over R, $S * G$ is the skew group algebra and $\Lambda_{\widetilde{Q}}$ is the preprojective algebra of a tame quiver $\widetilde{Q}$.

The algebraic Mckay correspondence-II

The correspondence is described by the diagram

$$R \underset{e(-)e}{\overset{\text{Auslander algebra}}{\rightleftarrows}} S * G \underset{\text{preprojective algebra}}{\overset{\text{degree 0 part}}{\rightleftarrows}} k\widetilde{Q} \xrightarrow{-/\langle e \rangle} kQ$$

where e comes from the projective $S * G$-module induced by R and Q is a Dynkin quiver. Furthermore, we have

$$\underline{MCM}(R) \cong \mathcal{D}^b(\mathbb{C}Q)/\tau \text{ (1-cluster category of } Q)$$